

Simon Wozny

curriculum vitæ

📄 <https://si8881wo.github.io>

Education and degrees

- Jul. 2020 – Jun. 2025 **PhD in Physics**, *Lund University/LTH*, Sweden.
thesis *Theory of transport in lowdimensional nanostructures*
thesis advisors Martin Leijnse (Lund University)
- Oct. 2018 – Jun. 2020 **Master of Science in Theoretical Physics**, *Lund University*, Sweden.
thesis *Investigation of gap properties in helical edge states of two-dimensional topological insulators*
supervisors Martin Leijnse (Lund University)
Sigurður I. Erlingsson (Reykjavik University)
- Oct. 2015 – Sept. 2018 **Bachelor of Science in Mathematics**, *University of Konstanz*, Germany.
thesis *Spektraltheoretische Störungssätze und Anwendung auf gyroskopische Systeme*
supervisor Robert Denk (University of Konstanz)
- Oct. 2012 – Mar. 2016 **Bachelor of Science in Physics**, *University of Konstanz*, Germany.
thesis *Influence of potential scattering of magnetic impurities in topological insulator edge states* (carried out at Reykjavik University, Iceland)
supervisors Wolfgang Belzig (University of Konstanz)
Sigurður I. Erlingsson (Reykjavik University)

Additional responsibilities

- Jun. 2023 – Jun. 2025 **NanoLund Student Representative**, *NanoLund Center for Nanoscience, Lund University*.
Board member of NanoLund, representing young researchers and organizing events aimed at them

Publications

- S. Wozny and Martin Leijnse, "Current noise in quantum dot thermoelectric engines", *Phys. Rev. B* 111, 075422 (2025).
doi:10.1103/PhysRevB.111.075422
- S. Matern, K. Macieszczak, S. Wozny and Martin Leijnse, "Metastability and quantum coherence-assisted sensing in interacting parallel quantum dots", *Phys. Rev. B* 107, 125424 (2023).
doi:10.1103/PhysRevB.107.125424
- S. Wozny, M. Leijnse and S. I. Erlingsson, "Dynamic impurities in two-dimensional topological-insulator edge states", *Phys. Rev. B* 104, 205418 (2021).
doi:10.1103/PhysRevB.104.205418
- S. Wozny, K. Vyborny, W. Belzig, and S. I. Erlingsson, "Gap formation in helical edge states with magnetic impurities", *Phys. Rev. B* 98, 165423 (2018).
doi:10.1103/PhysRevB.98.165423

Conference contributions

- August 2024 **HAMLET-Physics 2024 in Copenhagen.**
poster presentation: "Exploring transport in coupled quantum dots for quantum reservoir computing"
- March 2024 **DPG Spring meeting Berlin 2024.**
oral presentation in session TT 12: "Noise calculations for transport through quantum dot systems and applications in charge sensing and nano thermodynamics"
- September 2022 **DPG Spring meeting Regensburg 2022.**
oral presentation in session TT 18: "Dynamic impurities in two-dimensional topological insulator edge states"
- March 2020 **APS March meeting 2020, (online).**
oral presentation in session X45: "Dynamic impurities in two-dimensional topological insulator edge states"

Awards and prizes

- June 2025 **2025 NanoLund Award.**
Distinction for PhD students, for scientific breadth and for dedication to NanoLund research, education and outreach
- October 2022 **NanoLund Annual meeting 2022.**
Special unicorn prize, "Transport in Quantum Dots: Current, Noise and Thermodynamic Uncertainty Relations"
- August 2022 **COMPUTE summer retreat 2022.**
2nd place poster competition, "Transport in Quantum Dots: Current, Noise and Thermodynamic Uncertainty Relations"

International schools

- 11.-15. November 2024 **Chalmers School: Quantum thermodynamics meets quantum transport.**
including poster presentation, Gothenburg, Sweden
- 4.-9. June 2023 **Wilhelm and Else Heraeus School Open Quantum Systems and Mesoscopic Physics 2023.**
including poster presentation, Hyytiälä, Finland
- 7.-14. May 2023 **Capri spring school on transport in nanostructures 2023.**
including oral presentation, Capri, Italy
- 17.-20. August 2021 **PhD School of the Wallenberg Center of Quantum Technology.**
online, due to the covid-19 pandemic

Teaching Experience

- Spring 2025 **(Co)Supervision of MSc. Student, Lund University/LTH.**
- Winter 23/ Spring 24 **(Co)Supervision of undergraduate student projects, Lund University/LTH.**
- Winter 23/24 **Teaching assistant for "FAFA56 – Kvantfysikaliska Koncept", Lund University/LTH.**
weekly problem solving sessions and lab supervision
- Spring 2023 **Teaching assistant for "FFFA02 – Nanoscience and Nanotechnology - an Introduction", Lund University.**
project work supervision

- Spring/Winter 2023 **Teaching assistant for "FYB22/ÄFYD13 – Physics: Basic Quantum Mechanics", Lund University.**
lab supervision for the spectroscopy lab
- Spring 22/23 **Teaching assistant for "FAFA80 – Tillämpad vågrörelselära", Lund University/LTH.**
lab supervision for the diffraction lab
- Spring 2022 **(Co)Supervision of BSc. Student, Lund University/LTH.**
- Winter 20/21/22 **Teaching assistant for "FFN35/FYST24 – The Physics of Low-Dimensional Structures and Quantum Devices", Lund University/LTH.**
project work supervision
- Winter 20/21/22 **Teaching assistant for "FAFA55 – Kvantfysikaliska Koncept", Lund University/LTH.**
weekly problem solving sessions and lab supervision
- Spring 2021 **Teaching assistant for "FAFA75 - Energi- och omvärldsfysik", Lund University/LTH.**
lab supervision for the experimental methods lab
- Autumn 20/23 **Teaching assistant for "FAFA65/FAFA05 – Termodynamik, våglära och optik", Lund University/LTH.**
lab supervision for the experimental methods lab
- between 2013 – 2018 **Various mathematics and mathematics for physicists tutorials, University of Konstanz.**
correcting hand-ins and presenting solutions to weekly problem sets for groups of 5-15 students

Language Skills

- german native language
- english extensive knowledge
- swedish basic communication skills

References

References are available upon request